

T3 matched BSM EFT terms by field content

EFT order: 5; loop order: 1. Columns are determined only from matched BSM EFT terms. Empty cells denote no matched term with that field combination.

Field combinations 1 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H \ell$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\hbar B_{\mu_1 \mu_2}^2}{12\epsilon}$ $-\frac{1}{24}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)B_{\mu_1 \mu_2}^2$ $-\frac{1}{24}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right)B_{\mu_1 \mu_2}^2$	$+C_{t2}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{\text{lambda}d\text{a}H1\text{ln}v1^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{5\text{lambda}d\text{a}H1\text{ln}v2^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon}$ $+\frac{\text{lambda}d\text{a}H2\text{ln}v1^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{5\text{lambda}d\text{a}H2\text{ln}v2^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{6\epsilon}$ $+\frac{\text{lambda}d\text{a}H1\text{ln}v1\text{lambda}d\text{a}H1\text{ln}v2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{\text{lambda}d\text{a}H2\text{ln}v1\text{lambda}d\text{a}H2\text{ln}v2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{1}{2}\text{lambda}d\text{a}H1\text{ln}v1^2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{5}{6}\text{lambda}d\text{a}H1\text{ln}v2^2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{\text{lambda}d\text{a}H1\text{ln}v1\text{lambda}d\text{a}H1\text{ln}v2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}}$ $+\frac{1}{2}\text{lambda}d\text{a}H2\text{ln}v1^2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{5}{6}\text{lambda}d\text{a}H2\text{ln}v2^2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{\text{lambda}d\text{a}H2\text{ln}v1\text{lambda}d\text{a}H2\text{ln}v2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}}$ $+\frac{4\hbar\overline{\lambda_{\tau 3}}H_{a_1}H_{a_2}\lambda_{\tau 3}H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon}$ $+\frac{4\hbar\overline{\lambda_{\tau 3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\tau 3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{S_1}^2-M_{S_2}^2)}$ $+\frac{4\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)\overline{\lambda_{\tau 3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\tau 3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{S_1}^2-M_{S_2}^2)}$ $-\frac{4\hbar\overline{\lambda_{\tau 3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{\tau 3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{S_1}^2-M_{S_2}^2)}$ $-\frac{4\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right)\overline{\lambda_{\tau 3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{\tau 3}H_{a_1}^\dagger H_{a_2}^\dagger}{3(M_{S_1}^2-M_{S_2}^2)}$	$-\frac{\hbar\left(W_{\mu_1 \mu_2}^{A_1}\right)^2}{12\epsilon}$ $-\frac{1}{24}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)\left(W_{\mu_1 \mu_2}^{A_1}\right)^2$ $-\frac{1}{24}\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_2}^2}\right)\left(W_{\mu_1 \mu_2}^{A_1}\right)^2$	$+\frac{2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)\overline{W_{\tau 1, \tau 1}W_{\tau 2, \tau 2}}\ell_{a_1, a_2}{}^\tau C P_L \ell_{a_2, a_1}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\tau 1}\ell_{a_1, a_2}\ell_{a_3, a_4}M_F^2}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $-\frac{2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)\overline{W_{\tau 1, \tau 1}W_{\tau 2, \tau 2}}\ell_{a_1, a_2}{}^\tau C P_L \ell_{a_2, a_1}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\tau 1}\ell_{a_1, a_2}\ell_{a_3, a_4}M_F^2}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $+\frac{2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)\overline{W_{\tau 1}\ell_{a_3, a_4}}C P_R \overline{t_{a_3, a_4}}{}^\tau M_{S_1}^2W_{\tau 1, \tau 1}W_{\tau 2, \tau 2}\ell_{a_1, a_2}\ell_{a_3, a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $-\frac{2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)\overline{W_{\tau 1}\ell_{a_3, a_4}}C P_R \overline{t_{a_3, a_4}}{}^\tau M_{S_1}^2W_{\tau 1, \tau 1}W_{\tau 2, \tau 2}\ell_{a_1, a_2}\ell_{a_3, a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $+\frac{2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)\overline{W_{\tau 1, \tau 1}W_{\tau 2, \tau 2}}\ell_{a_4, a_2}{}^\tau C P_L \ell_{a_2, a_1}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\tau 1}\ell_{a_1, a_2}\ell_{a_3, a_4}M_F}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $-\frac{2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)\overline{W_{\tau 1, \tau 1}W_{\tau 2, \tau 2}}\ell_{a_4, a_2}{}^\tau C P_L \ell_{a_2, a_1}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\tau 1}\ell_{a_1, a_2}\ell_{a_3, a_4}M_F}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $+\frac{2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)\overline{\lambda_{\tau 1}\ell_{a_3, a_4}}C P_R \overline{t_{a_3, a_4}}{}^\tau M_{S_1}^2W_{\tau 1, \tau 1}W_{\tau 2, \tau 2}\ell_{a_1, a_2}\ell_{a_3, a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$ $-\frac{2\hbar \log\left(\frac{\mu_{\text{bar}}^2}{M_{S_1}^2}\right)\overline{\lambda_{\tau 1}\ell_{a_3, a_4}}C P_R \overline{t_{a_3, a_4}}{}^\tau M_{S_1}^2M_{S_1}^2W_{\tau 1, \tau 1}W_{\tau 2, \tau 2}\ell_{a_1, a_2}\ell_{a_3, a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\sqrt{3}\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H \ell$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3\hbar B_{\mu_1\mu_2}^2}{4\epsilon}$ $-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)B_{\mu_1\mu_2}^2$ $-\frac{1}{4}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2$	$+C_{\mathcal{H}2}H_{a_1}H_{a_1}^\dagger$ $-H_{a_1}\mu^2H_{a_1}^\dagger$ $+\frac{3\text{lambdaH1Inv}2^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{3\text{lambdaH2Inv}2^2\hbar H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon}$ $+\frac{3}{2}\text{lambdaH1Inv}2^2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{3}{2}\text{lambdaH2Inv}2^2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger$ $+\frac{2\hbar\overline{\lambda_{\text{T}3}}H_{a_1}H_{a_2}\lambda_{\text{T}3}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon}$ $+\frac{2\hbar\overline{\lambda_{\text{T}3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\text{T}3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$ $+\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\overline{\lambda_{\text{T}3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{\text{T}3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$ $-\frac{2\hbar\overline{\lambda_{\text{T}3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{\text{T}3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$ $-\frac{2\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\overline{\lambda_{\text{T}3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{\text{T}3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2}$	$-\frac{5\hbar\left(W_{\mu_1\mu_2}^A\right)^2}{6\epsilon}$ $-\frac{1}{2}\hbar\log\left(\frac{\mu\text{bar}2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^A\right)^2$ $-\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^A\right)^2$ $-\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^A\right)^2$	$\frac{2\sqrt{\frac{1}{2}}\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W_{\tau_1,\tau_1}W_{\tau_2,\tau_2}}\epsilon_{a_1,a_2}^T C P_L \epsilon_{a_2,\tau_1}H_{a_1}H_{a_2}M_{\text{H}}^2\lambda_{\text{T}1}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}M_F^2}{(M_F^2-M_{\text{H}}^2)(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)}$ $+\frac{2\sqrt{\frac{1}{2}}\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W_{\tau_1,\tau_1}W_{\tau_2,\tau_2}}\epsilon_{a_1,a_2}^T C P_L \epsilon_{a_2,\tau_1}H_{a_1}H_{a_2}M_{\text{H}}^2\lambda_{\text{T}1}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}M_F^2}{(M_F^2-M_{\text{H}}^2)(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)}$ $-\frac{2\sqrt{\frac{1}{2}}\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{\lambda_{\text{T}1}}\epsilon_{a_2,\tau_1}C P_R \epsilon_{a_1,a_2}^T M_{\text{H}}^2W_{\tau_1,\tau_1}W_{\tau_2,\tau_2}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{(M_F^2-M_{\text{H}}^2)(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)}$ $+\frac{2\sqrt{\frac{1}{2}}\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{\lambda_{\text{T}1}}\epsilon_{a_2,\tau_1}C P_R \epsilon_{a_1,a_2}^T M_{\text{H}}^2W_{\tau_1,\tau_1}W_{\tau_2,\tau_2}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{(M_F^2-M_{\text{H}}^2)(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)}$ $-\frac{2\sqrt{\frac{1}{2}}\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W_{\tau_1,\tau_1}W_{\tau_2,\tau_2}}\epsilon_{a_1,a_2}^T C P_L \epsilon_{a_2,\tau_1}H_{a_1}H_{a_2}M_{\text{H}}^2\lambda_{\text{T}1}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}M_F}{(M_F^2-M_{\text{H}}^2)(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)}$ $+\frac{2\sqrt{\frac{1}{2}}\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{W_{\tau_1,\tau_1}W_{\tau_2,\tau_2}}\epsilon_{a_1,a_2}^T C P_L \epsilon_{a_2,\tau_1}H_{a_1}H_{a_2}M_{\text{H}}^2\lambda_{\text{T}1}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}M_F}{(M_F^2-M_{\text{H}}^2)(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)}$ $-\frac{2\sqrt{\frac{1}{2}}\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{\lambda_{\text{T}1}}\epsilon_{a_2,\tau_1}C P_R \epsilon_{a_1,a_2}^T M_{\text{H}}^2M_{S_1}^2W_{\tau_1,\tau_1}W_{\tau_2,\tau_2}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2-M_{\text{H}}^2)(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)}$ $+\frac{2\sqrt{\frac{1}{2}}\hbar\log\left(\frac{\mu_F^2}{M_F^2}\right)\overline{\lambda_{\text{T}1}}\epsilon_{a_2,\tau_1}C P_R \epsilon_{a_1,a_2}^T M_{\text{H}}^2M_{S_1}^2W_{\tau_1,\tau_1}W_{\tau_2,\tau_2}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{(M_F^2-M_{\text{H}}^2)(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)}$

[illegible]

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	e
T3-A	0	1	0	3	1	2	$\frac{1}{2}$			

[illegible]

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell e$
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